

Cooperative Navigation for Maritime Autonomous Surface Ships through Communication-Enabled Distributed Intelligence

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Abstract

Cooperative navigation among Maritime Autonomous Surface Ships (MASS) is challenging in partially observable maritime environments, where limited local sensing restricts situational awareness during multi-vessel encounters. This paper develops a communication-enabled cooperative navigation framework for coordinated path planning and collision avoidance. Each vessel encodes its local observation into a compact latent message, while messages from multiple neighboring vessels are aggregated with its own observation to extend situational awareness beyond the local sensing range. A context-aware Mixture-of-Experts architecture identifies the dominant International Regulations for Preventing Collisions at Sea (COLREGs) obligation in multi-vessel encounters and guides the navigation policy accordingly. A three-stage curriculum progressively introduces individual navigation, inter-vessel communication, and COLREGs-compliant coordination. For large-scale validation under realistic vessel traffic, an AIS-driven digital twin is developed for a major maritime route across East and Southeast Asian waters, encompassing open-ocean, coastal, and strait environments. Large vessel populations derived from real AIS traffic are deployed across these environments to evaluate path planning and collision avoidance under diverse encounter conditions. The proposed framework improves cooperative navigation over conventional independent-navigation approaches, while t-SNE analysis reveals consistent structures in the learned communication space. The results demonstrate that communication-enabled distributed intelligence can support large-scale, COLREGs-compliant cooperative navigation under realistic maritime traffic.

Keywords: Maritime autonomous surface ships, Cooperative navigation, Inter-vessel communication, Collision avoidance, COLREGs compliance, AIS-driven digital twin

1. Introduction

The rapid advances of autonomous navigation technologies have brought Maritime Autonomous Surface Ships (MASS) to the forefront of maritime research and regulation. The International Maritime Organization (IMO) has examined the operation of MASS through its regulatory scoping exercise, reflecting growing interest in remotely controlled and fully autonomous vessels [?]. Despite this progress, safe and efficient navigation in congested waters remains a major challenge to the practical deployment of MASS [?]. Autonomous vessels need to avoid collisions while complying with the International Regulations for Preventing Collisions at Sea (COLREGs), particularly in dense maritime traffic where multiple vessels may interact simultaneously under uncertain and changing conditions [?].

Cooperative navigation involves multiple vessels coordinating their trajectories and sharing locally acquired information to improve navigational safety and efficiency [?]. Each vessel observes only part of the surrounding environment and exchanges locally available information with neighboring vessels. The re-

ceived information is combined with the vessel's own observation to determine its subsequent navigation behavior. Instead of relying on a complete global representation of the maritime environment, cooperative navigation distributes locally acquired information among vessels so that their trajectories can be coordinated according to surrounding traffic conditions.

Communication-enabled cooperative navigation involves three key operations. First, each vessel extracts navigation-relevant information from its local observation and encodes it into a compact message. Second, messages received from neighboring vessels within communication range are aggregated into a fixed-dimensional cooperative representation. Third, each vessel determines its trajectory by combining this representation with its own local observation. This information exchange extends situational awareness beyond individual sensing ranges and supports coordinated responses to potential conflicts [?]. This communication distinguishes cooperative navigation from independent collision avoidance, where surrounding vessels are treated as moving obstacles and each vessel determines its trajectory solely from locally available observations.

Classical path-planning techniques include graph-search algorithms such as A* and Dijkstra's algorithm, sampling-based planners, and artificial potential field methods (Zhang et al., 2021; García Maza and Argüelles, 2022) [?]. Graph-search

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methods can determine optimal or near-optimal routes with respect to a predefined discretization and cost function, whereas potential field methods provide computationally efficient local guidance. However, these methods rely on predefined representations of the navigation environment and do not directly apply for the coupled interactions among multiple autonomous vessels [?]. Model Predictive Control (MPC) incorporates vessel dynamics, operational constraints, and short-term motion predictions into online trajectory planning [?]. Its computational cost generally increases with the number of interacting vessels and the prediction horizon, which prevents from applying to complicated multi-vessel environments (Xue et al., 2024) [?]. Rule-based methods explicitly encode COLREGs requirements, but their navigation behavior is largely determined by predefined encounter categories and may become difficult to specify for multi-vessel encounter configurations [?].

Deep reinforcement learning (DRL) provides a learning-based approach to autonomous vessel navigation beyond predefined planning and encounter rules. Through repeated interactions with simulated maritime environments, an autonomous vessel can learn a collision-avoidance policy without explicitly programming every possible encounter and avoidance maneuver (Zhao and Roh, 2019; Li et al., 2021; Wu et al., 2023) [? ?]. Existing studies have shown that DRL can jointly account for goal reaching, collision avoidance, and COLREGs compliance within a learned navigation policy [?] [?]. However, most single-agent DRL approaches model surrounding vessels as dynamic obstacles rather than cooperative agents. These vessels neither exchange information nor coordinate their trajectories, leaving each agent to navigate solely from observations available within its own sensing range. This formulation limits situational awareness and does not support cooperative collision avoidance.

Cooperative navigation introduces additional learning challenges in multi-agent environments. As the policy of each vessel evolves during training, the transition dynamics observed by one agent change with the policies of other agents, creating a non-stationary learning environment that can destabilize joint policy learning (Wang and Zhao, 2025) [?]. Partial observability further limits the information available to each vessel [?]. Radar, cameras, and other onboard sensors have finite ranges and fields of view, and their measurements can be affected by occlusion, environmental conditions, and sensor uncertainty [?] [?]. As a result, each vessel determines its navigation behavior using incomplete information about the surrounding maritime traffic.

Inter-vessel communication can alleviate partial observability by exchanging information acquired locally by individual vessels [?]. Direct transmission of raw observations, however, increases communication overhead and may convey information irrelevant to the current navigation task. Learned communication instead encodes local observation of each vessel into a compact, task-relevant message optimized jointly with the navigation policy [? ?]. The exchange of these messages supplements locally available observations with information acquired by neighboring vessels and establish distributed situational awareness across the communicating vessels. Communi-

cation alone, however, does not ensure effective cooperation. Coordinated navigation depends on the information encoded in each message, the aggregation of messages from a varying number of neighbors, and the use of the aggregated information in selecting navigation actions. These factors become increasingly important in dense maritime traffic, where several vessels may simultaneously enter communication range and create interacting collision risks.

Multi-agent reinforcement learning (MARL) extends reinforcement learning to environments where multiple agents learn and act concurrently [?]. In cooperative MARL, agents learn policies that coordinate their actions toward a shared objective. This approach has been applied to cooperative collision avoidance, formation control, task allocation, and multi-vessel path planning [?] [?] [?]. In communication-enabled MARL, agents exchange observations or learned messages to obtain information unavailable from their local observations [?]. For example, Multi-Agent Deep Deterministic Policy Gradient (MADDPG) with explicit inter-agent communication can facilitate cooperative navigation and collision avoidance (Wang and Zhao) [?]. Their agents establish a consistent communication protocol and used shared observations to compensate for missing local information. However, this study is limited to two-vessel environments. The aggregation of messages from multiple simultaneous neighbors, which becomes essential when several vessels are within communication range, has not been explicitly addressed.

Multi-agent reinforcement learning (MARL) extends reinforcement learning to environments where multiple agents learn and act concurrently [?]. In cooperative MARL, agents learn policies that coordinate their actions toward a shared objective, with applications including cooperative collision avoidance, formation control [?], task allocation [?], and multi-vessel path planning [?]. Communication-enabled MARL further incorporates the exchange of observations or learned messages to supplement information unavailable from local observations [?]. For example, Multi-Agent Deep Deterministic Policy Gradient MADDPG with explicit inter-agent communication facilitate cooperative navigation and collision avoidance (Wang and Zhao) [?]. Their agents establish a consistent communication protocol and used shared observations to compensate for missing local information. However, existing studies have largely focused on small-scale multi-vessel settings, including two-vessel environments. They have not sufficiently examined dense maritime traffic where multiple vessels simultaneously exchange information and coordinate collision-avoidance maneuvers. When a vessel receives messages from several neighbors simultaneously, the messages vary in number and order. Direct concatenation produces a policy input so that its dimension depends on the number of neighboring vessels and its representation changes with their ordering. Simple aggregation may instead suppress information from a neighbor posing an immediate collision risk [?]. A permutation-invariant aggregation mechanism is introduced to transform a variable number of messages into a fixed-dimensional cooperative representation while preserving navigation-relevant information.

COLREGs compliance presents another challenge in multi-

vessel environments. A vessel may simultaneously act as the give-way vessel relative to one neighbor and the stand-on vessel relative to another [? 26]. Treating each pairwise encounter independently and averaging the corresponding compliance rewards may obscure the obligation associated with the most immediate collision risk. Cooperative navigation in such encounters needs to account for both shared observations and the relative priority of concurrent COLREGs obligations. Jointly learning goal-directed navigation, collision avoidance, inter-vessel communication, and COLREGs compliance further increases the difficulty of policy learning. When these objectives are introduced simultaneously, agents learn basic navigation while interpreting messages generated by evolving policies and responding to encounter-dependent maritime rules. The resulting coupling can destabilize training and hinder the development of meaningful communication. Curriculum learning provides a way to reduce this coupling by introducing navigation objectives progressively according to their complexity [? 25].

To address these challenges, a communication-enabled framework is developed for cooperative MASS navigation under partial observability. Each vessel encodes its local observation into a compact latent message and exchanges it with neighboring vessels. Messages received from multiple neighbors are aggregated into a fixed-dimensional cooperative representation, which is combined with local observation of individual vessels to determine its navigation action. The communication range extends beyond the local observation range, which provides information about neighboring vessels before they become locally observable and supporting earlier coordination during potential collision encounters.

Multi-vessel COLREGs encounters are handled through a context-aware Mixture-of-Experts (MoE) architecture. Rule-specific experts evaluate individual COLREGs obligations, while a gating network identifies the obligation calling for the most immediate response and determines the corresponding compliance reward. This prioritization avoids averaging compliance rewards across concurrent pairwise encounters with different levels of collision risk. A progressive three-stage curriculum introduces individual navigation, inter-vessel communication, and COLREGs-compliant coordination sequentially. Each vessel first learns goal-directed navigation and collision avoidance, then learns to exchange and use information from neighboring vessels, and finally refines its cooperative navigation according to COLREGs. Vessels exchange learned messages over a predefined communication range that extends beyond their local observation range. Information from neighboring vessels can thus be received before those vessels become locally observable. Each message provides a compact representation of transmitting vessel’s local observation and supplements the information available to the receiving vessel. This extended situational awareness supports earlier recognition of potential conflicts and coordinated avoidance maneuvers, reducing dependence on short-range reactive responses.

Large-scale experiments evaluate the proposed framework using an AIS-driven digital twin of a major maritime route across East and Southeast Asian waters. The digital twin reproduces real-world vessel traffic in open-ocean, coastal, and strait

environments, covering different traffic densities and encounter patterns. Large vessel populations create simultaneous multi-vessel interactions that extend beyond the small-scale settings commonly examined in existing studies. The experiments examine cooperative path planning and collision avoidance under realistic maritime traffic and assess how inter-vessel communication supports navigation as the number of interacting vessels increases.

The main contributions of this paper are summarized as follows:

- **Communication-enabled cooperative navigation under partial observability:** A learned inter-vessel communication mechanism combines compact latent message exchange with multi-neighbor aggregation. A variable number of neighboring messages are transformed into a fixed-dimensional cooperative representation and combined with local observations, extending situational awareness beyond individual sensing ranges without assuming complete global information.
- **Context-aware prioritization of COLREGs obligations:** A MoE-based reward structure addresses concurrent COLREGs obligations in multi-vessel encounters. Its rule-conditioned gating network identifies the obligation calling for the most immediate response and determines the corresponding compliance reward, preventing critical obligations from being diluted by averaging across pairwise encounters.
- **Progressive learning of cooperative navigation:** A three-stage curriculum introduces individual navigation, inter-vessel communication, and COLREGs-compliant coordination sequentially. This structure separates the acquisition of basic navigation behavior from communication and rule-compliant coordination, reducing the difficulty of learning these coupled objectives simultaneously.
- **Large-scale validation under realistic maritime traffic:** An AIS-driven digital twin of a major maritime route across East and Southeast Asian waters is developed to evaluate the proposed framework across open-ocean, coastal, and strait environments. Large vessel populations derived from real AIS traffic provide diverse traffic densities and encounter patterns for examining cooperative path planning and collision avoidance beyond small-scale encounter configurations.

The remainder of this paper is organized as follows. Section 2 introduces the vessel dynamics and COLREGs requirements. Section 3 reviews path-planning, cooperative navigation, and communication-enabled learning methods for MASS. Section 4 presents the detailed description of the proposed cooperative navigation framework. Section 5 presents the simulation environment and experimental settings. Section 6 discusses the experimental results. Section 7 reports the ablation studies, and Section 8 concludes the paper.

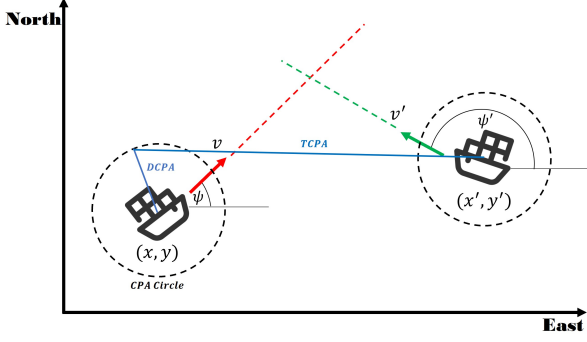


Figure 1: Vessel control model for collision avoidance event.

2. Background

This section describes the vessel motion model and the COLREGs considered for autonomous navigation. The motion model characterizes the kinematic and dynamic behavior of a vessel, while the COLREGs specify the navigation rules for collision avoidance in different encounter situations.

2.1. Vessel Dynamics

Fig. 1 shows a vessel control model. A vessel controls its motion through propulsion thrust and rudder angle, represented by the control vector τ . The vessel pose in the earth-fixed frame is defined as $\eta = [x, y, \psi]^T$, where x and y denote the position and ψ denotes the heading angle. The body-fixed velocity vector is defined as $v = [u, v, r]^T$, where u , v , and r denote surge velocity, sway velocity, and yaw rate, respectively. The body-fixed velocity is transformed into the earth-fixed frame by the rotation matrix

$$\mathbf{T}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (1)$$

The vessel kinematics are then expressed as

$$\dot{\eta} = \mathbf{T}(\psi)v, \quad (2)$$

where the body-fixed velocity determines the evolution of the vessel position and heading in the earth-fixed frame.

The translational and rotational motion of the vessel is governed by

$$\mathbf{M}\dot{v} + \mathbf{C}(v)v + \mathbf{D}(v)v + \mathbf{g}(\eta) = \mathbf{B}\tau, \quad (3)$$

where $\mathbf{M}$ is the inertia matrix including rigid-body and added-mass terms, $\mathbf{C}(v)$ represents Coriolis and centripetal effects, $\mathbf{D}(v)$ represents hydrodynamic damping, $\mathbf{g}(\eta)$ represents hydrostatic restoring forces, and $\mathbf{B}$ maps the control inputs to the forces and moments acting on the vessel. Vessel motion is restricted to the horizontal plane, with the sway velocity set to $v = 0$. Under this assumption, the motion is represented by surge and yaw dynamics, and the hydrodynamic effects in (3) are characterized using acceleration, deceleration, and drag coefficients. This reduced representation describes the longitudinal motion and heading changes considered in the navigation environment.

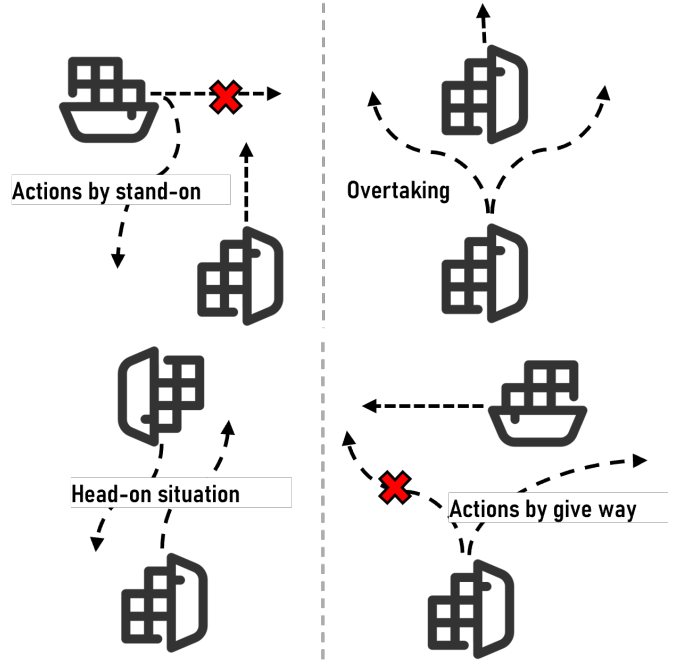


Figure 2: Encounter situations described by COLREGs.

2.2. Collision Avoidance and COLREGs

The International Regulations for Preventing Collisions at Sea (COLREGs) specify navigation rules for preventing collisions between vessels. Among these regulations, Rules 13-17 define the principal obligations associated with the encounter situations considered in this study. These situations are illustrated in Fig. 2 and summarized below.

- **Overtaking (Rule 13):** An overtaking vessel keeps clear of the vessel being overtaken until it is finally past and clear.
- **Head-On (Rule 14):** When two vessels approach on reciprocal or nearly reciprocal courses, each vessel alters its course to starboard so that they pass on the port side of each other.
- **Crossing (Rule 15):** In a crossing encounter, the vessel that has the other vessel on its starboard side acts as the give-way vessel and keeps clear.
- **Give-Way Vessel (Rule 16):** The give-way vessel takes early and substantial action to keep clear of the other vessel.
- **Stand-On Vessel (Rule 17):** The stand-on vessel maintains its course and speed but may take action to avoid collision when the give-way vessel does not take appropriate action.

These rules specify the navigation obligations associated with the pairwise encounter types considered in this study. In a multi-vessel environment, however, a vessel may simultaneously encounter several neighboring vessels and hold different

COLREGs obligations relative to each vessel. For example, the same vessel may act as a give-way vessel relative to one neighbor while acting as a stand-on vessel relative to another.

Collision risk between two vessels can be characterized using the Distance at Closest Point of Approach (DCPA) and Time to Closest Point of Approach (TCPA). DCPA represents the predicted minimum separation between two vessels if their current courses and speeds are maintained, while TCPA represents the time remaining until the closest point of approach. A small DCPA indicates a close predicted encounter, while a small positive TCPA indicates that the closest approach will occur in the near future. These quantities jointly characterize the spatial and temporal proximity of an encounter and are commonly used for collision-risk assessment [10].

3. Related Work

Related studies on MASS navigation can be broadly grouped into path planning and cooperative navigation. Path-planning studies focus on generating feasible routes and collision-avoidance trajectories for individual vessels, while cooperative-navigation studies consider how multiple vessels exchange information and coordinate their trajectories. This section reviews both research directions and identifies the limitations addressed by the proposed framework.

3.1. Path Planning for MASS

MASS path planning determines a sequence of positions and control actions that guide a vessel from its origin to a destination while satisfying navigational constraints. Global path planning identifies waypoints that define the overall route from the origin to the destination, whereas local path planning determines detailed trajectories between consecutive waypoints by accounting for real-time vessel dynamics, surrounding traffic, and collision risks [16].

Global path-planning methods such as A*, Dijkstra’s algorithm, and geodesic-based routing can determine optimal or near-optimal routes when an accurate map of the navigation environment is available [13]. A finer map representation improves route planning but increases the search space and computational cost. These methods also assume that the environment remains largely unchanged during planning, making precomputed routes less suitable when navigation conditions change. Data-driven approaches have used Automatic Identification System (AIS) records to incorporate historical vessel movements and environmental conditions into route planning. Neural-network models have generated fuel-efficient routes in Arctic waters using historical speed, course, and ice-concentration data [22]. LSTM-based models have also predicted intermediate waypoints from vessel turning-point data [23]. These methods use historical navigation data to improve route generation, but they do not directly respond to collision risks that emerge during navigation.

Local path planning addresses short-term changes by determining detailed trajectories between consecutive waypoints according to nearby traffic and evolving encounter conditions.

Artificial potential fields, the dynamic window approach, and model predictive control have been widely studied for real-time collision avoidance [13, 17]. Their computational burden can increase as the number of interacting vessels grows. Rule-based methods provide explicit collision-avoidance actions for predefined encounter conditions but become difficult to specify when vessel behaviors and multi-vessel configurations vary. DRL provides a learning-based approach in which collision-avoidance behavior is acquired by maximizing cumulative rewards through repeated interactions with simulated maritime environments. Existing studies incorporate goal reaching, collision avoidance, and COLREGs compliance into the reward structure. Encounter classifiers have been used to identify the applicable COLREGs situation and activate the corresponding rule-specific reward terms [10]. Observation structures have also been developed for varying numbers of surrounding vessels, including fixed-length distance records and spatial partitioning based on COLREGs encounter sectors [28, 10]. Despite these advances, existing DRL approaches generally determine collision-avoidance actions independently for each vessel. Surrounding vessels are incorporated into local observations, but information acquired by different vessels is not exchanged to supplement their limited observations. As a result, each vessel plans its trajectory from locally available information without coordinating its avoidance maneuver with neighboring vessels.

3.2. Cooperative MASS Navigation

When vessels navigate independently, information about neighboring vessels is obtained primarily through onboard sensing and externally available traffic information such as AIS data. Although these sources provide observed motion states, they provide limited information about the future navigation intentions of neighboring vessels. Predicting how surrounding vessels will maneuver during an evolving encounter is difficult. Cooperative navigation supplements this information through direct inter-vessel communication, such as AIS, Very High Frequency (VHF) radio. Vessels can exchange their current states together with intended courses and planned maneuvers, providing neighboring vessels with information that cannot be inferred reliably from observations alone. Such information sharing improves situational awareness and supports coordinated collision avoidance [2, 11].

Communication strategies for cooperative MASS navigation differ in the information exchanged among vessels. Existing studies have considered positional information and collision-risk measures such as CPA [13], control intentions including rudder commands and speed changes [14], and historical trajectories for predicting the future motion of neighboring vessels [12]. More recent MARL approaches encode local observations into compact latent messages learned jointly with the navigation policy. These messages provide task-relevant representations of locally acquired information that neighboring agents can incorporate into their own observations [32].

Wang and Zhao [32] have applied explicit message passing to cooperative vessel navigation using MADDPG and reported improved collision-avoidance performance over single-

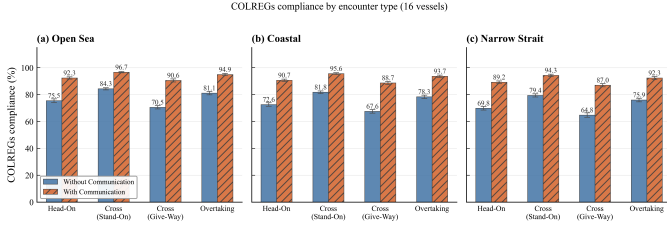


Figure 3: Overview of the proposed system across three nested spatial scales. (A) The global origin-to-destination voyage against real-world AIS traffic. (B) Decomposition of the route into a sequence of local planning regions joined by entry/exit hand-off points. (C) The cooperative path-planning task solved within a single region, which is the focus of this paper.

agent baselines. However, the study considers a two-vessel environment, where each vessel receives information from only one neighboring vessel. In a multi-vessel environment, a vessel may receive messages from several neighbors simultaneously. This creates additional problem of combining a varying number of messages into a consistent policy input. Pairwise treatment of COLREGs encounters also becomes insufficient when a vessel simultaneously holds different navigation obligations relative to multiple neighboring vessels. Furthermore, existing communication-enabled learning studies have primarily been demonstrated in small-scale multi-vessel settings, leaving cooperative navigation with large vessel populations under realistic maritime traffic insufficiently examined.

To this end, the proposed framework addresses these limitations through multi-neighbor message aggregation and context-aware prioritization of concurrent COLREGs obligations. A progressive three-stage curriculum introduces individual navigation, inter-vessel communication, and COLREGs-compliant coordination sequentially. Large-scale experiments further examine the resulting cooperative navigation behavior using an AIS-driven digital twin covering open-ocean, coastal, and strait environments.

4. Proposed Framework

This section presents the proposed framework for cooperative MASS navigation. The overall navigation setting and multi-agent formulation are first introduced, followed by the observation representation, inter-vessel message passing, COLREGs assessment, and three-phase curriculum used to learn cooperative navigation.

4.1. System Overview

Fig. 3 illustrates the navigation setting across three spatial scales. Global scale (Fig. 3A) describes the complete origin-to-destination voyage over a large maritime area. At this scale, the route may extend over several thousand kilometers and pass through maritime environments with substantially different traffic characteristics. This type of voyage covers approximately 4,606 km across East and Southeast Asian waters, corresponding to about 6.1 days at a service speed of 17 knots. The route includes open-ocean, coastal, and strait environments, and

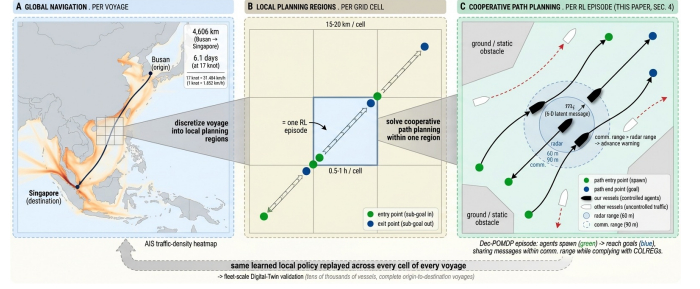


Figure 4: collaborative navigation framework

real-world AIS data are used to represent the vessel traffic encountered along the voyage. The global route is divided into consecutive regions over which navigation conditions can be handled over shorter distances and time horizons (Fig. ??B). Each region covers approximately 15-20 km, corresponding to about 0.5-1 h of transit. A vessel enters a region through a designated entry point and proceeds toward its corresponding exit point, which connects the current region to the subsequent one. This decomposition provides a sequence of local navigation tasks while preserving progression toward the final destination. Local scale describes the cooperative navigation task considered by the proposed framework. Multiple vessels within the same region navigate toward their respective exits while responding to surrounding traffic and obstacles. Each vessel makes short-range navigation adjustments according to its local observation and information received from neighboring vessels (Fig. 3C). The radar and communication ranges are set to 60 m and 90 m, respectively. The larger communication range provides information from communicating vessels before they enter the local radar range, supporting earlier coordination of collision-avoidance maneuvers. Cooperative path planning, collision avoidance, and COLREGs compliance are jointly handled at this scale.

Consider N vessel agents indexed by $\mathcal{I} = \{1, \dots, N\}$. Let $\mathcal{S}$ denote the global state space, $\mathcal{A}$ the joint action space, and $\mathcal{O}$ the observation space. The system state evolves according to the transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$, and the corresponding reward is determined by $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. The observation function $\Omega : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{O}$ specifies the information available to the agents, and $\gamma \in (0, 1)$ denotes the discount factor. Based on these elements, the cooperative MASS navigation problem is formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) defined by $\langle \mathcal{I}, \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{O}, \Omega, \gamma \rangle$. Each agent i has access to its local observation $o_i \in \mathcal{O}$ and messages received from neighboring agents. The communication neighbors of agent i are defined as

$$\mathcal{N}_i = \{j \neq i : d(i, j) \leq r_{\text{comm}}\}, \quad (4)$$

where $d(i, j)$ denotes the distance between agents i and j , and r_{comm} is the communication range. Each neighboring agent $j \in \mathcal{N}_i$ generates a latent message $\mathbf{m}_j$ from its local observation. Agent i combines the received messages into the aggregated message $\bar{\mathbf{m}}_i$, which supplements its local observation for cooperative navigation.

Fig. 5 illustrates the overall network and training architecture. Each agent employs a MessageActor π_{Θ_M} and a ControlActor π_{Θ_C} , with their parameters shared across the vessel agents. The MessageActor generates the latent message transmitted to neighboring agents, while the ControlActor determines the navigation action from the local observation and aggregated neighbor message $\bar{\mathbf{m}}_i$. The Critic also incorporates $\bar{\mathbf{m}}_i$ during training, and an auxiliary COLREGs classifier provides an additional training signal for encounter classification.

The framework follows centralized training with decentralized execution (CTDE). During training, the actor and Critic networks are updated using the information available to the training architecture. During execution, each agent independently generates messages and determines its navigation action using only its local observation and messages received from neighboring agents. The same learned policy can be executed independently by each vessel without access to the global state.

4.2. Observation Model

At each timestep, agent i constructs its local observation from four components: radar measurements, goal information, its own dynamic state, and the current COLREGs encounter type. The radar observation $\mathbf{O}_{\text{radar},i} \in \mathbb{R}^{360}$ represents a 360° omnidirectional scan with an angular resolution of 1° . For each ray angle $\theta \in \{0^\circ, 1^\circ, \dots, 359^\circ\}$, the radar measurement is normalized as

$$O_{\text{radar},i}[\theta] = \frac{d_i(\theta)}{r_{\text{radar}}} - 0.5, \quad (5)$$

where $d_i(\theta)$ is the distance from agent i to the nearest obstacle along direction θ , and r_{radar} is the radar detection range. The normalized measurement lies in $[-0.5, 0.5]$, with 0.5 assigned when no obstacle is detected within the radar range. The goal observation $\mathbf{O}_{\text{goal},i} \in \mathbb{R}^2$ represents the destination relative to the current vessel position in ego-centric polar coordinates:

$$\mathbf{O}_{\text{goal},i} = \left[\frac{d_{g,i}}{d_{\text{max}}}, \frac{\theta_{g,i}}{180^\circ} \right]. \quad (6)$$

The distance to the goal is given by

$$d_{g,i} = \|\mathbf{x}_{\text{goal},i} - \mathbf{x}_i\|, \quad (7)$$

where d_{max} is the map diagonal used for normalization. The relative goal direction is defined as

$$\theta_{g,i} = \text{SignedAngle}(\mathbf{f}_i, \mathbf{x}_{\text{goal},i} - \mathbf{x}_i), \quad (8)$$

where $\mathbf{f}_i$ denotes the forward direction of vessel i and $\theta_{g,i} \in (-180^\circ, 180^\circ]$. The self-state observation $\mathbf{O}_{\text{self},i} \in \mathbb{R}^4$ contains the vessel speed, yaw rate, heading angle, and rudder angle:

$$\mathbf{O}_{\text{self},i} = \left[\frac{v_i^{\text{spd}}}{v_{\text{max}}}, \frac{r_i}{r_{\text{max}}}, \frac{\psi_i}{180^\circ}, \frac{\delta_i}{\delta_{\text{max}}} \right]. \quad (9)$$

Here, v_i^{spd} denotes the vessel speed, r_i the yaw rate, ψ_i the heading angle, and δ_i the rudder angle. Each quantity is normalized to either $[-1, 1]$ or $[0, 1]$ according to its physical range. The

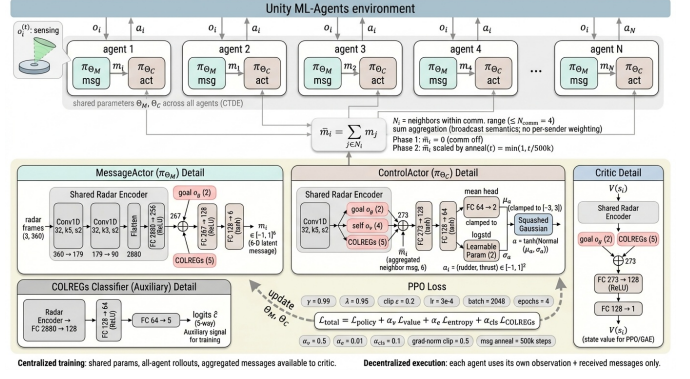


Figure 5: Message-passing architecture for cooperative navigation.

COLREGs observation $\mathbf{O}_{\text{COLREGs},i} \in \mathbb{R}^5$ represents the dominant encounter type c_i^* identified by the COLREGs assessor. The set of encounter types is defined as

$$\mathcal{C} = \{\text{None}, \text{HeadOn}, \text{CrossStandOn}, \text{CrossGiveWay}, \text{Overtaking}\}. \quad (10)$$

The dominant encounter type satisfies $c_i^* \in \mathcal{C}$. Its one-hot representation is given by

$$\mathbf{O}_{\text{COLREGs},i} = \text{onehot}(c_i^*) \in \{0, 1\}^5, \quad (11)$$

with $\|\mathbf{O}_{\text{COLREGs},i}\|_1 = 1$.

The complete local observation is obtained by concatenating the four components:

$$\mathbf{o}_i = [\mathbf{O}_{\text{radar},i} \parallel \mathbf{O}_{\text{goal},i} \parallel \mathbf{O}_{\text{self},i} \parallel \mathbf{O}_{\text{COLREGs},i}]. \quad (12)$$

The resulting network input has 371 dimensions. The absolute position (x_i, z_i) is additionally provided to the training environment only to identify communication neighbors. It is not included in $\mathbf{o}_i$ or used as an input to the policy networks. The environment therefore maintains 373 values per agent, while the policy networks receive only the 371-dimensional local observation.

4.3. Message passing

Fig. 5 illustrates the message-passing architecture used to exchange locally acquired information among neighboring vessels. Each agent first encodes its local observation into a compact latent message. Messages received from multiple communication neighbors are aggregated and combined with the local observation of the receiving agent to determine its control action.

The MessageActor π_{Θ_M} encodes the local observation of agent i into a six-dimensional latent message $\mathbf{m}_i \in [-1, 1]^6$. The stacked radar frames are first processed by two one-dimensional convolutional layers to extract spatial features. The radar embedding is then combined with the goal, self-state, and COLREGs observations to form a message given by

$$\mathbf{h}_i = [\mathbf{f}_{\text{conv}}(\mathbf{O}_{\text{radar},i}) \parallel \mathbf{O}_{\text{goal},i} \parallel \mathbf{O}_{\text{self},i} \parallel \mathbf{O}_{\text{COLREGs},i}]. \quad (13)$$

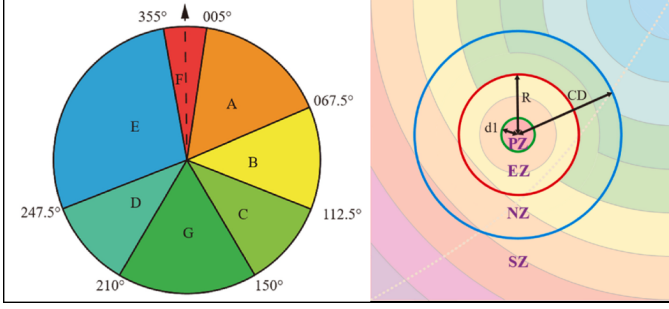


Figure 6: COLREGs encounter types considered by the rule-specific experts.

The latent message is generated through a fully connected layer with a tanh activation given as

$$\mathbf{m}_i = \tanh(\mathbf{W}_{\text{msg}} \mathbf{h}_i + \mathbf{b}_{\text{msg}}). \quad (14)$$

Each agent broadcasts its latent message to the neighboring agents within the communication range. Agent i receives messages from the agents in $\mathcal{N}_i$ and aggregates them as

$$\bar{\mathbf{m}}_i = \sum_{j \in \mathcal{N}_i} \mathbf{m}_j. \quad (15)$$

This aggregation converts messages from a varying number of communication neighbors into a fixed six-dimensional representation. Information from all communicating neighbors can be incorporated without changing the input dimension as the number of neighboring vessels varies.

The aggregated message $\bar{\mathbf{m}}_i$ is combined with the local features of agent i and provided to the *ControlActor* π_{Θ_c} . As shown in Fig. 5, the radar embedding, goal observation, self-state observation, COLREGs observation, and aggregated message form a 273-dimensional input. Two fully connected layers reduce this representation from 273 to 128 and then to 64 dimensions using tanh activations. The *ControlActor* generates the continuous control action

$$\mathbf{a}_i = (\delta_i, \tau_i) \in [-1, 1]^2, \quad (16)$$

where δ_i and τ_i represent the normalized rudder and thrust commands, respectively. A squashed Gaussian policy is used to produce bounded continuous actions. The pre-squash action is sampled according to $\tilde{\mathbf{a}}_i \sim \mathcal{N}(\boldsymbol{\mu}_{a,i}, \boldsymbol{\sigma}_a)$, and the control action is obtained as

$$\mathbf{a}_i = \tanh(\tilde{\mathbf{a}}_i). \quad (17)$$

The corresponding log-probability is corrected for the tanh transformation as

$$\log \pi(\mathbf{a}_i | \mathbf{s}_i) = \log p(\tilde{\mathbf{a}}_i | \mathbf{s}_i) - \sum_k \log(1 - a_{i,k}^2 + \varepsilon), \quad (18)$$

where ε is a small constant for numerical stability, and $\boldsymbol{\sigma}_a$ denotes the learnable action standard deviation.

4.4. COLREGs Situation Assessment via Mixture of Experts

In a multi-vessel environment, agent i may simultaneously encounter several vessels and hold different COLREGs obligations relative to each of them. Fig. 6 illustrates the four encounter types considered in the proposed framework: head-on (H), crossing stand-on (S), crossing give-way (G), and overtaking (O). The proposed Mixture-of-Experts (MoE) assessor evaluates these pairwise encounters and identifies the obligation calling for the most immediate response. Each encounter type is associated with a rule-specific expert, while a fuzzy urgency measure determines which expert governs the current COLREGs assessment. For each neighboring vessel j within the radar range, the pairwise encounter type $c_i(j)$ is first identified from the relative geometry between vessels i and j as

$$c_i(j) = \begin{cases} \text{HeadOn}, & \text{if } |\phi_{ij}| < 15^\circ, |\phi_{ji}| < 15^\circ, \\ \text{Overtaking}, & \text{if } |\phi_{ji}| > 112.5^\circ, v_i > 1.1v_j, \\ \text{CrossGiveWay}, & \text{if } 5^\circ < \phi_{ij} < 112.5^\circ, \\ \text{CrossStandOn}, & \text{if } -112.5^\circ < \phi_{ij} < -5^\circ, \\ \text{None}, & \text{otherwise.} \end{cases} \quad (19)$$

Here, ϕ_{ij} is the signed bearing from agent i to vessel j relative to the heading of agent i , with positive angles indicating the starboard side. Similarly, ϕ_{ji} denotes the corresponding bearing from vessel j to agent i .

The urgency of each pairwise encounter is evaluated using three quantities. Let d_j denote the current distance from agent i to vessel j . The Time to Closest Point of Approach (TCPA), denoted by τ_j , represents the remaining time until the two vessels reach their closest point under their current motions. The Distance at Closest Point of Approach (DCPA), denoted by δ_j , represents the corresponding minimum separation distance. Each quantity is mapped to a normalized urgency measure in $[0, 1]$. The distance-based urgency is defined as

$$\mu_d(d_j) = 1 - \frac{d_j}{r_{\text{radar}}}, \quad (20)$$

where r_{radar} is the radar detection range. Thus, the urgency increases as vessel j approaches agent i . The TCPA-based urgency is defined as

$$\mu_\tau(\tau_j) = \frac{1}{1 + \tau_j/T_0}, \quad (21)$$

where $T_0 = 60$ s is the reference time. A shorter TCPA produces a larger urgency value. The DCPA-based urgency is defined as

$$\mu_\delta(\delta_j) = \max\left(0, 1 - \frac{\delta_j}{D_0}\right), \quad (22)$$

where $D_0 = 50$ m is the reference distance. A smaller predicted separation produces a larger urgency value, while the DCPA contribution becomes zero when $\delta_j \geq D_0$. These urgency measures are combined to quantify the overall urgency of the encounter with vessel j . The encounter urgency is defined as

$$g_i^j = \beta_{c_i(j)} \left[w_d \mu_d(d_j) + w_\tau \mu_\tau(\tau_j) + w_\delta \mu_\delta(\delta_j) \right], \quad (23)$$

where w_d , w_τ , and w_δ determine the relative contributions of distance, TCPA, and DCPA, respectively, and satisfy

$$w_d + w_\tau + w_\delta = 1. \quad (24)$$

The weights are set to $w_d = 0.3$, $w_\tau = 0.4$, and $w_\delta = 0.3$. The encounter-dependent coefficient $\beta_{c_i(j)}$ assigns different urgency scales to the four encounter types:

$$\beta_{c_i(j)} = \begin{cases} 2.0, & c_i(j) = \text{HeadOn}, \\ 1.5, & c_i(j) = \text{CrossGiveWay}, \\ 1.3, & c_i(j) = \text{CrossStandOn}, \\ 1.2, & c_i(j) = \text{Overtaking}. \end{cases} \quad (25)$$

After evaluating all pairwise encounters within the radar range, the neighbor associated with the highest urgency is selected as

$$j_i^* = \arg \max_{j \in \mathcal{N}_i^{\text{radar}}} g_i^j, \quad (26)$$

where $\mathcal{N}_i^{\text{radar}}$ denotes the set of neighboring vessels detected by the radar of agent i . The corresponding dominant COLREGs encounter is then given by $c_i^* = c_i(j_i^*)$. If $g_i^{j_i^*} < g_{\min}$, no active encounter is declared at the current timestep and the COLREGs reward is suppressed. The dominant encounter type c_i^* serves two roles. First, it is encoded as the one-hot COLREGs observation $\mathbf{O}_{\text{COLREGs},i}$ defined in the observation model. Second, it selects the rule-specific compliance criterion used to evaluate the current navigation behavior. The resulting compliance score $r_{c_i^*} \in [-2, 2]$ reflects the current heading and speed of agent i under the selected COLREGs obligation. Thus, only the obligation associated with the dominant encounter contributes to the COLREGs reward at each timestep, instead of averaging compliance scores across simultaneous pairwise encounters.

The total reward for agent i is defined as

$$R_i = R_{\text{goal},i} + R_{\text{collision},i} + R_{\text{spinning},i} + R_{\text{time},i} + R_{\text{forward},i} + R_{\text{COLREGs},i}. \quad (27)$$

The goal reward $R_{\text{goal},i} = 100$ is assigned when the vessel arrives within the threshold distance d_{arrive} of its destination. A collision produces $R_{\text{collision},i} = -100$ and immediately terminates the episode. The spinning penalty $R_{\text{spinning},i} = -80$ terminates the episode when the accumulated net yaw in one direction exceeds 180° . A time penalty $R_{\text{time},i} = -0.1$ is applied at each timestep. Forward motion is rewarded according to

$$R_{\text{forward},i} = 0.1 \frac{v_i}{v_{\max}}. \quad (28)$$

The rule-specific compliance score is incorporated into the COLREGs reward as

$$R_{\text{COLREGs},i} = \alpha_c r_{c_i^*} (1 + g_i^{j_i^*}), \quad (29)$$

where α_c controls the contribution of COLREGs compliance to the total reward. It is set to 0.30 in Phase 1 and increased to 0.45 in Phase 3. The factor $(1 + g_i^{j_i^*})$ increases the influence of the compliance score as the urgency of the dominant encounter increases.

In addition to the rule-specific reward, an auxiliary classifier is trained to predict the dominant COLREGs encounter type from the radar observation. The classification loss is defined as

$$\mathcal{L}_{\text{COLREGs}} = - \sum_{c \in \mathcal{C}} y_{i,c} \log \hat{p}_{i,c}, \quad (30)$$

where $y_{i,c}$ is the one-hot target determined by c_i^* and $\hat{p}_{i,c}$ is the predicted probability of encounter type c . The classification loss is incorporated into the overall training objective as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{policy}} + \alpha_v \mathcal{L}_{\text{value}} + \alpha_e \mathcal{L}_{\text{entropy}} + \alpha_{\text{cls}} \mathcal{L}_{\text{COLREGs}}, \quad (31)$$

where the loss weights are set to $\alpha_v = 0.5$, $\alpha_e = 0.01$, and $\alpha_{\text{cls}} = 0.1$. As shown in Fig. 5, the classifier provides an auxiliary training signal from the radar observation and does not directly select the control action during execution.

4.5. Three-phase curriculum training

Policy training becomes more difficult when new cooperative behaviors are introduced before basic navigation behavior has been sufficiently established. In particular, the communication policy initially produces messages with little useful information, while the control policy must simultaneously learn how to interpret these messages and respond to COLREGs obligations. A three-phase curriculum is introduced to separate these learning processes. Phase 1 establishes individual navigation, Phase 2 introduces inter-vessel communication, and Phase 3 strengthens COLREGs compliance after cooperative navigation has been learned.

In Phase 1, inter-vessel communication is disabled by setting all incoming message vectors to zero. Each agent therefore learns to navigate using only its local observation. The training establishes goal-directed navigation and collision avoidance while discouraging inefficient behaviors such as excessive turning. The COLREGs reward is included with its initial coefficient $\alpha_c = 0.30$, providing a basic level of rule compliance without communication. The resulting policy parameters provide the initial navigation capability for the next phase.

In Phase 2, inter-vessel communication is activated while the policy parameters learned in Phase 1 are used for initialization. The MessageActor begins encoding local observations into latent messages, and messages received from neighboring vessels are aggregated and provided to the ControlActor. The agents consequently learn how information unavailable from their own observations can be incorporated into navigation decisions. This phase extends the individual navigation behavior established in Phase 1 to cooperative navigation based on inter-vessel information exchange.

In Phase 3, the COLREGs reward coefficient is increased from $\alpha_c = 0.30$ to $\alpha_c = 0.45$. At this stage, the agents already use exchanged messages for cooperative navigation, allowing training to place greater emphasis on encounter-specific COLREGs obligations. The increased reward weight refines the navigation policy toward COLREGs-compliant cooperative maneuvers without introducing communication and stronger rule compliance simultaneously.

Fig. 7 shows the training progression across the three phases. The transition to Phase 2 initially changes the accumulated reward as the agents begin to generate and interpret latent messages. The subsequent training integrates this exchanged information into the navigation policy. Phase 3 then shifts greater emphasis toward COLREGs compliance through the increased reward coefficient. The curriculum thereby develops the policy progressively from individual navigation to communication-assisted cooperation and finally to reinforced COLREGs-compliant cooperative navigation.

5. Simulation Environment

The proposed framework is implemented in a maritime simulation environment that reproduces vessel motion, local sensing, inter-vessel communication, and COLREGs encounters. The environment is used for policy training and controlled evaluation, while the large-scale AIS-driven digital twin introduced later extends the evaluation to realistic vessel traffic across open-ocean, coastal, and strait environments. This section describes the simulation platform, vessel configuration, sensing and communication models, and training parameters.

5.1. Simulation Platform

The simulation environment is implemented in Unity 3D and connected to a Python-based training pipeline through the ML-Agents toolkit. Vessel motion, surrounding obstacles, onboard sensing, and inter-vessel interactions are reproduced in Unity, while policy training is performed through the Python pipeline. No field experiments with physical vessels are conducted; all training and evaluation are performed in simulation, including the large-scale AIS-driven digital-twin experiments described later. Table 1 summarizes the principal environment and agent parameters.

During training, four vessel agents are initialized at randomized positions selected from predefined spawn points. Each vessel is assigned a distinct goal located at least 50 m from its initial position. The initial heading is directed toward the assigned goal with a random perturbation within $\pm 10^\circ$, producing variations in encounter geometry across episodes. Vessel motion is further diversified by scaling the maximum speed of each agent by a factor sampled from $\mathcal{U}(0.8, 1.8)$. The initial speed is then set between 20% and 50% of the resulting maximum speed.

Each vessel is equipped with an omnidirectional radar represented by 360 rays at 1° angular resolution and a detection range of 60 m. The normalized radar measurements from three consecutive frames are stacked to capture short-term changes in the surrounding environment. In addition to the radar measurements, each agent observes goal-relative distance and bearing, its kinematic state consisting of speed, yaw rate, heading, and rudder angle, and the one-hot COLREGs encounter representation. These components form the 371-dimensional input to the policy network. Two additional dimensions containing the absolute vessel position are used only to determine communication neighbors and are not provided to the policy network, resulting in a 373-dimensional observation maintained by the

simulation environment. The continuous action space has two dimensions corresponding to the rudder and thrust commands.

For inter-vessel communication, each vessel transmits a six-dimensional latent message to neighboring vessels within a communication range of 90 m. Up to four nearest communication partners are considered at each timestep. The communication range is intentionally larger than the 60 m radar range, allowing information from neighboring vessels to be received before they become directly observable through the local radar. The message generation and aggregation mechanism is described in Section 4.3.

Three maritime environments are constructed to evaluate the navigation policy under different operating conditions: *Open Ocean*, *Narrow Channel*, and *Coastal*. The *Open Ocean* environment represents relatively unconstrained navigation in open water, where vessels have greater maneuvering space and encounters are comparatively sparse. The *Narrow Channel* environment constrains the navigable area and consequently produces more frequent close-range encounters with limited maneuvering space. The *Coastal* environment represents denser multi-vessel traffic in waters affected by surrounding geographical constraints. All three environments use the same vessel dynamics, sensing model, communication mechanism, and policy architecture. Their spawn-point geometry and navigational constraints are varied to reproduce the different traffic and encounter conditions. The three environments therefore provide controlled settings for examining how the learned cooperative navigation policy responds to changes in traffic density, encounter frequency, and available maneuvering space.

An episode terminates upon goal arrival, collision, excessive spinning, or when the maximum episode length is reached. Goal arrival is declared when the vessel approaches its assigned destination within $d_{\text{arrive}} = 15$ m. A collision terminates the episode immediately, while excessive spinning is detected when the accumulated net yaw in one direction exceeds 180° . Episodes without an earlier terminal event are limited to 10,000 simulation steps. The trained models and simulation environment are publicly available.¹

5.2. Vessel Dynamics

Each vessel is modeled as a rigid body moving on the horizontal plane, with pitch, roll, and heave excluded from the simulation. Longitudinal motion is determined by the thrust command together with the acceleration, deceleration, and drag parameters, while the rudder command controls the change in heading. The corresponding vessel parameters are listed in Table 1. The nominal maximum speed is 5.0 m/s. To introduce differences in vessel motion, the maximum speed of each agent is scaled at initialization by a factor sampled from $\mathcal{U}(0.8, 1.8)$. The same dynamic model is used in the *Open Ocean*, *Narrow Channel*, and *Coastal* environments. Differences among the three environments therefore originate from their spawn-point geometries and navigational constraints instead of changes in the vessel dynamics. The simulation time scale is set to 20 $\times$ during training to accelerate experience collection.

¹https://github.com/DragonTrainerTristana/DT_Vessel

Table 1: Simulation environment and vessel parameters.

Parameter	Symbol	Value
Vessel mass (normalized)	m	1.0
Vessel length	L	10.0 m
Vessel beam	B	2.0 m
Maximum speed	$v_{\max}$	5.0 m/s
Maximum turn rate	$\delta_{\max}$	30°/s
Acceleration rate	a_{acc}	0.5 m/s ²
Deceleration rate	a_{dec}	0.2 m/s ²
Drag coefficient	C_d	0.1
Spawn area	—	400 × 400 m
Agents per training episode	—	4
Goal arrival threshold	d_{arrive}	15 m
Min. spawn-to-goal distance	—	50 m
Maximum episode length	$T_{\max}$	10,000 steps
Simulation time scale	—	20×

5.3. Radar and Communication Model

Each vessel is equipped with an omnidirectional radar sensor implemented using 360 raycasts at 1° intervals and a height of 1 m above the waterline. The radar detection range is set to $r_{\text{radar}} = 60$ m. The resulting distance measurements provide the local obstacle and vessel information described in the observation model. Three consecutive radar frames are stacked to capture short-term changes in the surrounding environment.

Inter-vessel information exchange in practical maritime environments can be supported by several communication technologies with different coverage, latency, and bandwidth characteristics. Wi-Fi provides low-latency and high-rate communication over relatively short distances, making it suitable mainly for port and near-shore operations. Very High Frequency (VHF) radio provides longer-range ship-to-ship communication and is widely used for maritime safety services, although its limited bandwidth restricts the amount of information that can be exchanged. Automatic Identification System (AIS), operating over maritime VHF channels, broadcasts vessel information such as position, speed, and course and can provide information about surrounding vessels beyond the range of onboard sensing.

Broadband cellular networks provide another communication option in coastal waters. LTE and 5G offer lower latency and higher data rates than conventional maritime VHF communication, but their availability depends on terrestrial network coverage and consequently decreases as vessels move farther offshore. Satellite communication extends connectivity to offshore and open-ocean environments. Traditional geostationary satellite systems provide wide-area coverage but introduce relatively large propagation delays. Low Earth Orbit (LEO) satellite systems reduce this propagation distance and provide lower communication latency, making satellite connectivity more suitable for time-sensitive maritime information exchange.

The proposed framework does not reproduce the physical-layer characteristics of a particular communication technology. Instead, inter-vessel communication is represented by a range-limited broadcast model that captures the availability of navigation information beyond the local radar range. A vessel can

Table 2: PPO training hyperparameters and network configuration.

Hyperparameter	Symbol	Value
Discount factor	γ	0.99
GAE lambda	λ	0.95
Learning rate	α	3×10^{-4}
Batch size	B	2,048
PPO epochs per update	K	4
PPO clipping	ϵ	0.2
Entropy bonus coefficient	α_e	0.01
Value loss coefficient	α_v	0.5
COLREGs coefficient (Phase 1)	α_c	0.30
COLREGs coefficient (Phase 3)	α_c	0.45
Classifier loss weight	α_{cls}	0.10
Gradient clipping norm	—	0.5
Phase 1 training steps	—	4.07×10^6
Total training steps	—	30×10^6
Observation dimension	—	371
Message dimension	$ \mathbf{m} $	6
Frame stacking	F	3
Conv. kernel sizes	—	5, 3
Conv. channels	—	32, 32
FC hidden units	—	256–128–64

exchange messages with neighboring vessels within $r_{\text{comm}} = 90$ m, while the radar detection range is $r_{\text{radar}} = 60$ m. The communication range is intentionally set larger than the radar detection range: $r_{\text{comm}} > r_{\text{radar}}$. This range difference makes information from a communicating vessel available before that vessel enters the local radar range. It represents the advance information that maritime communication systems, such as AIS or other ship-to-ship links, can provide beyond onboard sensing.

Each communicating vessel broadcasts the six-dimensional latent message generated by its *MessageActor*. For agent i , the set of available communication neighbors is

$$\mathcal{N}_i = \{j \neq i : d_{ij} \leq r_{\text{comm}}\}, \quad (32)$$

where d_{ij} denotes the distance between vessels i and j . Messages received from these vessels are combined according to the aggregation mechanism described in Section 4.3. When more than $N_{\text{comm}} = 4$ vessels are within communication range, the four nearest vessels are selected as communication partners. This constraint limits the number of simultaneously processed messages while preserving multi-neighbor information exchange in dense encounters.

5.4. Training Hyperparameters

The policy is trained using Proximal Policy Optimization (PPO). Table 8 summarizes the optimization parameters and network configuration used throughout the three-phase curriculum. The network architecture remains unchanged across the three phases, while the availability of inter-vessel communication and the COLREGs reward coefficient are varied according to the curriculum. Phase 1 comprises 4.07×10^6 training steps, and the complete curriculum comprises 30×10^6 steps.



Figure 7: Training reward curves for the three curriculum phases. Phase 1 (blue): no communication, baseline COLREGs weight. Phase 2 (red): communication ON, COLREGs 1 $\times$. Phase 3 (green): communication ON, COLREGs 1.5 $\times$. Shaded regions indicate ± 1 standard deviation.

6. Results and Discussion

This section evaluates the proposed framework along several complementary axes: training convergence across the three curriculum phases; scalability with respect to fleet size; a qualitative trajectory study; the structure of the learned message space; and a detailed, trajectory-level comparison of behavior across the three navigational settings. All evaluations are conducted with 16 vessels over 30,000 simulation steps unless otherwise noted.

6.1. Training Convergence

Fig. 7 shows the average reward curves for all three training phases. Phase 1 (blue, no communication, baseline COLREGs weight) converges to an average reward of approximately 2.1 by 4×10^6 steps, establishing a stable single-vessel navigation policy. At the Phase 2 transition (red, communication ON, COLREGs 1 $\times$), a transient reward drop is observed as agents begin processing the initially uninformative incoming messages; the curve recovers and surpasses the Phase 1 asymptote within approximately $1\text{--}2 \times 10^6$ additional steps. Phase 3 (green, communication ON, COLREGs 1.5 $\times$) begins at approximately 10×10^6 steps and achieves the highest converged reward of approximately 2.4, demonstrating that the elevated COLREGs coefficient further improves performance once cooperative communication is already established.

The strict ordering Phase 1 < Phase 2 < Phase 3 beyond the recovery period confirms that both inter-agent communication and stronger COLREGs enforcement independently contribute to navigation performance, and that the three-phase curriculum successfully introduces each component without catastrophic forgetting.

6.2. Scalability

Fig. 8 evaluates performance as fleet size increases from 4 to 16 vessels in the Open Ocean environment. For the non-communicating baseline, collision rate grows steeply ($0.3 \rightarrow 0.8 \rightarrow 1.6$), COLREGs compliance degrades ($88.5\% \rightarrow 83.2\% \rightarrow 79.2\%$), and goal success declines ($94.2\% \rightarrow 89.5\% \rightarrow 85.4\%$), reflecting the combinatorial growth of pairwise encounters with fleet density.

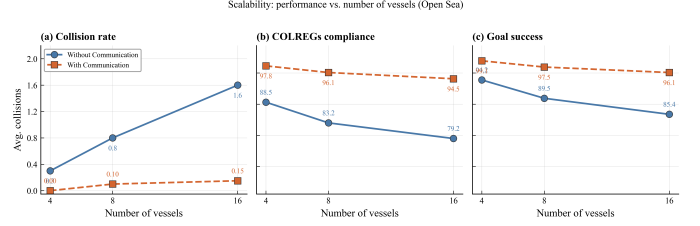


Figure 8: Scalability analysis in Open Ocean as fleet size varies from 4 to 16 vessels. (a) Collision rate. (b) COLREGs compliance (%). (c) Goal success rate (%).

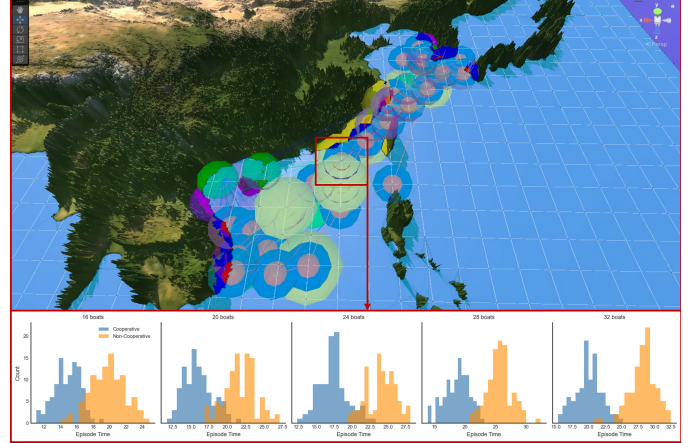


Figure 9: Episode time distributions for cooperative (blue) and non-cooperative (orange) agents at fleet sizes of 16–32 vessels. The cooperative distribution is consistently shifted toward shorter episode times and shows lower variance.

The communication-enabled agent is substantially more resilient: collision rate remains near zero across all fleet sizes ($\approx 0.0 \rightarrow 0.10 \rightarrow 0.15$), COLREGs compliance stays above 94% ($97.8\% \rightarrow 96.1\% \rightarrow 94.5\%$), and goal success remains above 96% ($\approx 94.7\% \rightarrow 97.5\% \rightarrow 96.1\%$). The near-flat performance curves demonstrate that sum-aggregated message passing with bounded neighborhood size $N_{\text{comm}} = 4$ scales gracefully as fleet density increases.

6.3. Qualitative Trajectory Analysis

Fig. 10 compares OS trajectories in a representative crossing encounter under identical initial conditions with and without message passing. Without communication, the OS detects the nearest target at 45 m (within radar range) at step 102 and executes a pronounced avoidance loop, yielding a trajectory length of 227 m. With communication enabled, the OS receives message-encoded state information from a vessel at 83 m — beyond the 60 m radar horizon — at step 100, triggering an earlier and shallower course adjustment. The resulting trajectory (218 m) bypasses the loop entirely. The overlay plot (Fig. 11) quantifies the behavioral divergence: at step 152, the two trajectories reach a maximum spatial separation of 49 m, demonstrating the advance-warning benefit of the extended communication range.

6.4. Message Space Analysis

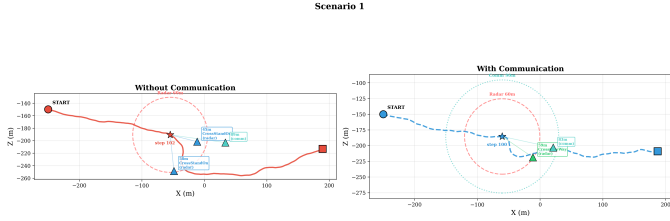


Figure 10: Scenario 1 trajectory comparison. Left: without communication — the OS executes a large avoidance loop after late radar detection at 45 m (step 102). Right: with communication — the OS receives advance state information from a vessel at 83 m (step 100) and completes a smoother, shorter path.

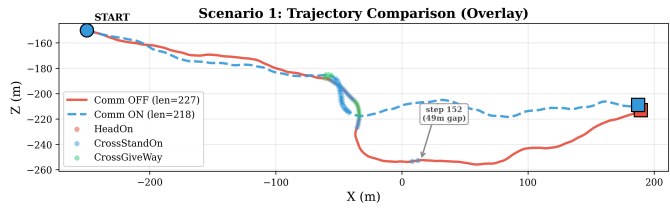


Figure 11: Trajectory overlay for Scenario 1. Comm OFF (red solid, length 227 m) vs. Comm ON (blue dashed, length 218 m). The maximum spatial divergence of 49 m occurs at step 152.

To verify that the learned 6-dimensional messages encode semantically meaningful navigation states, we apply t-SNE dimensionality reduction to raw observation vectors and generated message vectors collected during evaluation episodes. Fig. 12 presents eight subplots.

The 371-dimensional observation space (Fig. 12a) exhibits substantial cluster overlap, particularly between CrossStandOn and CrossGiveWay, indicating that raw radar observations do not cleanly separate encounter types. In contrast, the 6-dimensional message space (Fig. 12b) achieves markedly cleaner separation despite the 60-fold dimensionality reduction. Restricting to encounter-only timesteps (Fig. 12c) reveals four tightly separated clusters corresponding to the four active encounter types, confirming that the MessageActor compresses situation-discriminative information while suppressing irrelevant sensor noise.

Coloring by continuous variables exposes further interpretable structure. CrossGiveWay points exhibit predominantly positive yaw rates (Fig. 12d, warm colors), consistent with Rule 16’s requirement to turn to starboard. Goal distance (Fig. 12e) follows a smooth gradient within each cluster, indicating the message jointly encodes encounter situation and goal proximity. Speed (Fig. 12f) and goal angle (Fig. 12g) show similar cluster-aligned gradients. The encounter-only yaw rate panel (Fig. 12h) confirms the behavioral asymmetry between CrossGiveWay (positive yaw) and CrossStandOn (near-zero yaw) agents, directly reflecting their respective COLREGs obligations.

6.5. Behavior Across Distinct Navigational Settings

The three evaluation environments are not arranged along a single axis of increasing difficulty; each is designed to isolate

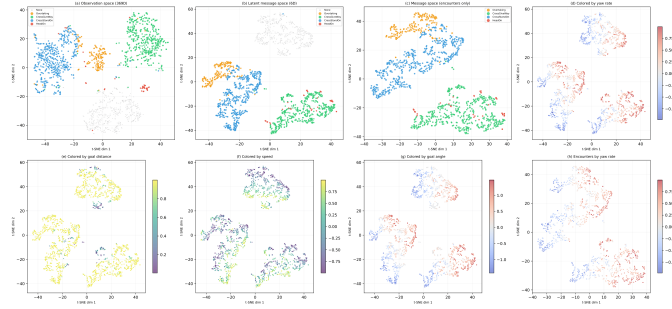


Figure 12: t-SNE analysis of the learned message space. (a) 371-dimensional observation space colored by COLREGs situation. (b) 6-dimensional message space, all timesteps. (c) Message space, encounter timesteps only. (d)–(h) Message space colored by yaw rate, goal distance, speed, goal angle, and encounter-only yaw rate respectively.

a qualitatively different kind of navigational challenge. Open Ocean presents unobstructed water in which interaction with other vessels constitutes essentially the entire task. The Narrow Channel concentrates traffic into limited lateral space, so that vessels routinely travel and pass one another at close range. The Coastal setting introduces static terrain—coastline and shoals—that must be skirted continually while moving traffic is simultaneously present. Read through this lens, the results below show that communication does not merely “help more when conditions worsen”; rather, it contributes in a distinct manner suited to what each setting demands. Throughout, we report a finer trajectory-level analysis over 10 independent runs of 2,000 steps, comparing the communication-disabled (COMM OFF) and communication-enabled (COMM ON) agents across the four panels of Fig. 13: per-rule COLREGs compliance (Fig. 13a), mean DCPA (Fig. 13b), dangerous-proximity frequency (Fig. 13c), and two efficiency measures—a control-effort proxy for fuel use (Fig. 13d) and time-to-goal (Fig. 13e). The control-effort proxy is defined as the per-step Euclidean norm of the commanded thrust and rudder [33]; smaller values indicate smoother, less actuator-intensive maneuvering and, by extension, lower fuel use.

Open Ocean: pure inter-vessel coordination.. With no terrain to contend with, behavior in open water is governed almost entirely by how vessels coordinate with one another. Most metrics are already favorable without communication: encounters are infrequent—an average of only 0.37 vessels within 50 m (Fig. 13c)—and the geometric margin is comfortable, with a mean DCPA of 49.88 m (Fig. 13b). Enabling communication leaves these essentially unchanged: DCPA edges to 50.61 m (Fig. 13b), the control-effort proxy from 0.89 to 0.88 (Fig. 13d), and time-to-goal from 162 to 161 steps (Fig. 13e). This near-invariance is precisely what makes open water valuable as a diagnostic: it isolates the one aspect of coordination that communication is decisive for even when space is abundant—the stand-on obligation. A vessel required by Rule 17 to *hold* its course and speed cedes the initiative to the give-way vessel; but an agent that cannot perceive its neighbor’s intent has no way to tell a situation in which it should hold from one in which it must act, and therefore defaults to a self-protective evasive maneuver

that violates the rule. Communication removes this ambiguity, producing the single largest effect in the entire study: stand-on compliance rises from 34.5% to 86.0% (Fig. 13a). The remaining rules, which an agent can satisfy from its own geometry, are already high and change little—Overtaking $\approx 99\%$, Head-On $84.1 \rightarrow 86.6$, and Give-Way $80.2 \rightarrow 87.7$ (all Fig. 13a). Open Ocean thus offers the cleanest evidence that communication’s primary role is to resolve intent ambiguity between vessels, independent of any spatial pressure.

Narrow Channel: coordination under close spacing.. A narrow channel changes the character of the task. Lateral room is limited, so vessels proceed and pass in sustained close company—a defining feature of the setting rather than a sign of failure. This signature is directly visible in the geometry: the mean DCPA is intrinsically smaller, at 24.85 m (Fig. 13b), and vessels share close quarters far more often, an average of 1.77 within 50 m—the highest of any environment (Fig. 13c). Close, continuous passing is exactly the regime in which shared intent is most useful, and it is here that communication delivers its largest absolute reduction in dangerous proximity, from 1.77 to 1.50 nearby vessels (Fig. 13c). Because tight spacing leaves little room to absorb a late reaction, the advance intent exchanged between neighbors converts into steadier passing rather than abrupt correction: the control-effort proxy falls from 0.82 to 0.75 (Fig. 13d) and time-to-goal shortens from 184 to 177 steps (Fig. 13e), while the stand-on recovery observed in open water persists, $31.8 \rightarrow 77.9\%$ (Fig. 13a). Where Open Ocean asked whether vessels coordinate at all, the narrow channel asks whether they can coordinate while packed closely together—and the cooperative agents demonstrably can.

Coastal: sharing capacity between traffic and terrain.. The coastal setting adds a second, qualitatively different obligation: vessels must continually avoid static terrain—coastline and shoals—at the same time as other traffic. An agent here must divide its maneuvering between two distinct demands, and because the navigable water is itself constricted by the surrounding geography, vessels are naturally drawn into close passing, with a mean DCPA of just 8.15 m (Fig. 13b). Communication’s contribution in this regime is best understood as *offloading the vessel-vessel half of the problem*: by receiving neighbors’ intent, an agent resolves traffic conflicts with smaller and earlier adjustments, freeing maneuvering capacity that can be redirected toward terrain. The consequences are visible across every panel. The safety margin shows its largest relative improvement of any environment, mean DCPA rising from 8.15 to 11.14 m (+37%, Fig. 13b); the efficiency gains are likewise the largest, with the control-effort proxy falling from 0.77 to 0.66 (Fig. 13d) and time-to-goal from 204 to 197 steps (Fig. 13e); dangerous-proximity events fall from 1.23 to 1.06 (Fig. 13c); and stand-on compliance again recovers strongly, $30.0 \rightarrow 72.5\%$ (Fig. 13a). Consistent with the interpretation that freed capacity is redirected to terrain, the cooperative agent also maintains greater clearance from static obstacles than its non-cooperative counterpart. The coastal results indicate that communication relieves the vessel-vessel burden a non-cooperative agent must other-

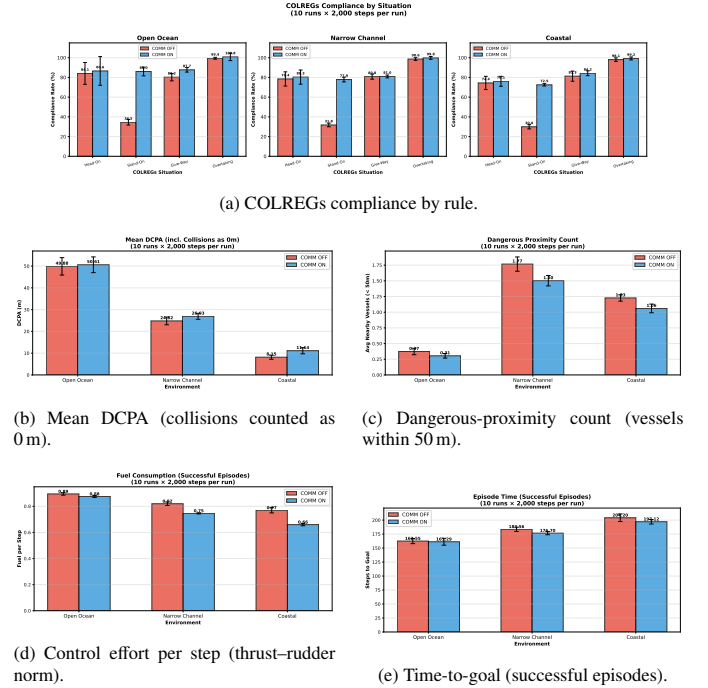


Figure 13: Trajectory-level behavior in the three navigational settings— Open Ocean (pure inter-vessel coordination), Narrow Channel (close-spacing coordination), and Coastal (traffic and terrain combined). 16 vessels, 10 runs $\times$ 2,000 steps; COMM OFF (red) versus COMM ON (blue). (a) Per-rule COLREGs compliance, showing the large stand-on (Rule 17) recovery under communication. (b) Mean DCPA. (c) Dangerous-proximity count. (d) Control-effort proxy for fuel use. (e) Time-to-goal.

wise absorb, allowing that capacity to be spent on the terrain-avoidance demands unique to this setting.

Control behavior.. The aggregate control-effort reductions reported above manifest, at the level of an individual encounter, as visibly smoother steering. Fig. 14 contrasts the rudder-angle trace over the first 20 steps of a representative encounter: the cooperative agent (blue) produces low-amplitude commands, transitioning gradually from a slight port correction to a moderate starboard bearing, whereas the non-cooperative agent (orange) exhibits larger, more abrupt deflections reaching approximately $\pm 20^\circ$. This contrast is the trajectory-level signature of the same advance-warning mechanism: an agent that receives a neighbor’s intent before radar contact can begin a gentle correction early, rather than reacting sharply once the encounter is already developing.

7. Ablation Studies

The results of Section 6 establish that the complete framework substantially outperforms a non-cooperative agent. This section isolates the contribution of each individual design choice. We first define a baseline that removes all of our contributions, then ablate, in turn, the message-aggregation scheme, the message dimensionality, the communicating-vessel proportion, the Mixture-of-Experts (MoE) reward architecture, and the three-phase curriculum. Unless stated otherwise, all ablations

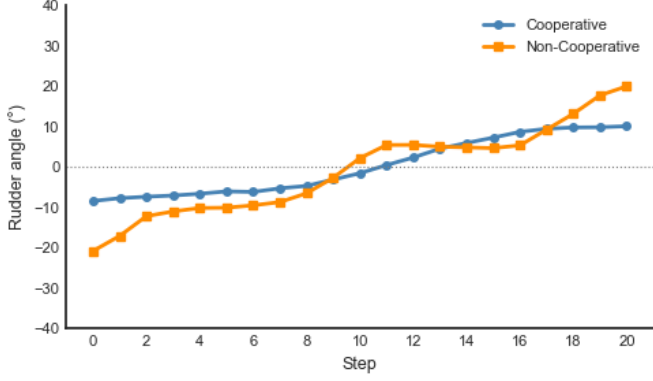


Figure 14: Rudder angle over the first 20 steps of a representative encounter. The cooperative agent (blue) produces smoother, smaller corrections; the non-cooperative agent (orange) exhibits larger reactive deflections.

are conducted in the Open Ocean environment with 16 vessels, averaged over 10 independent runs of 2,000 steps each, and report mean $\pm$ standard deviation. Table 3 summarizes the five studies at a glance; the subsections that follow present each in detail.

7.1. Baseline Definition

To quantify the cumulative effect of our contributions, we construct a baseline that strips the proposed framework down to its essential cooperative-communication core. Concretely, this configuration replaces multi-neighbor sum aggregation with single-neighbor (nearest-1) messaging, replaces the per-rule MoE experts with a single shared policy network, and replaces the staged curriculum with joint optimization of all objectives from the first training step. This reduced configuration reproduces the core design of the state-of-the-art cooperative message-passing approach of Wang and Zhao [32], and we therefore adopt it as our primary baseline. We emphasize that our contributions are evaluated not against a weakened strawman but against the prevailing design for cooperative MASS communication; moreover, by reconstructing that design within our own large-scale Digital-Twin environment rather than the original two-vessel particle setting, we obtain a controlled, scale-matched comparison that the original formulation could not provide. Each subsection below re-introduces one component on top of this baseline and measures the resulting change in collision rate, COLREGs compliance, goal-success rate, and navigation efficiency.

7.2. Message Aggregation: Sum vs. Nearest-Neighbor

The first and most fundamental design question is how an agent should combine messages when several neighbors are simultaneously within communication range. We compare our sum-of-neighbors aggregation against the nearest-1 scheme of the baseline, holding all other components fixed (message dimension $|\mathbf{m}| = 6$, identical training budget). Fig. 15 reports the comparison.

Sum aggregation eliminates collisions entirely in the evaluation episodes (0.00 vs. 0.80 collisions per run for nearest-1),

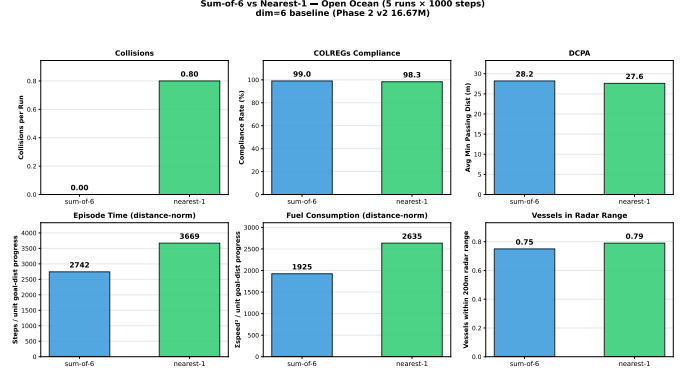


Figure 15: Message-aggregation ablation in Open Ocean (16 vessels, 5 runs $\times$ 1,000 steps, $|\mathbf{m}| = 6$). Sum-of-neighbors aggregation (blue) versus nearest-1 messaging (green) across collisions, goal success, COLREGs compliance, DCPA, episode time, and fuel consumption.

while also reducing distance-normalized episode time by approximately 25% (2,742 vs. 3,669) and fuel consumption by approximately 27% (1,925 vs. 2,635). COLREGs compliance and mean DCPA are comparable between the two schemes (99.0% vs. 98.3%; 28.2 m vs. 27.6 m), indicating that the safety margin per encounter is similar but that the nearest-1 agent resolves multi-vessel situations far less efficiently. This is the expected failure mode of single-neighbor messaging: by attending to only one partner at a time, the agent remains blind to all other vessels in range and must therefore react reactively as each new neighbor becomes the nearest, producing longer and more circuitous trajectories. Sum aggregation, in contrast, lets a vessel condition its action on the superposition of all in-range neighbors at once, which is precisely the broadcast semantics of VHF/AIS that the scheme is designed to emulate. This result validates multi-neighbor aggregation as a prerequisite—rather than an incremental refinement—for cooperative navigation beyond the two-vessel case.

7.3. Message Dimensionality

We next study how the capacity of the latent message—its dimensionality $|\mathbf{m}|$ —affects performance, sweeping $|\mathbf{m}| \in \{2, 4, 6, 8, 10, 12\}$ while keeping the architecture and training budget fixed. Fig. 16 reports COLREGs compliance and episode time as a function of message dimension.

Both metrics are essentially flat for $|\mathbf{m}| \geq 6$: compliance plateaus near 84% (84.3% at $|\mathbf{m}| = 6$, with no measurable gain at 8, 10, or 12) and episode time saturates near 238 steps. Dimensions below 6 (2 and 4) are slightly worse and exhibit higher run-to-run variance, indicating that very low-dimensional messages cannot fully encode the situation-discriminative information the receiver requires. The learning curves (Fig. 17) show that larger dimensions converge faster early in training, but all dimensions ≥ 6 reach an indistinguishable asymptotic reward. Message dimension was therefore selected as the smallest value that reaches the performance plateau: larger messages accelerate early convergence but yield no asymptotic benefit, while incurring greater bandwidth cost—a consideration that matters directly for the VHF/AIS channels

Table 3: Summary of ablation studies. Each row isolates one design choice against the baseline configuration of Section 7.1. Arrows indicate the direction of improvement attributable to the proposed choice.

Ablated component	Compared conditions	Principal finding
Message aggregation	Sum-of-neighbors vs. nearest-1	Collisions $0.80 \rightarrow 0.00$; episode time -25% ; fuel -27% .
Message dimension	$ \mathbf{m} \in \{2, \dots, 12\}$	Performance saturates at $ \mathbf{m} = 6$; larger dimensions add bandwidth cost without asymptotic gain.
Communicating proportion	8R/8C $\rightarrow$ 2R/14C; per-type split	Fleet compliance rises monotonically with adoption; communicating agents dominate receive-only agents at every mix.
MoE reward architecture	Specialized experts vs. capacity-matched shared net (gating fixed)	Small, consistent gain concentrated in the hardest rules (Crossing Stand-On); no capacity increase.
Three-phase curriculum	Phase 1 vs. Phase 2 vs. Phase 3; joint baseline	Strict ordering Phase 1 < Phase 2 < Phase 3; staging avoids the instability of simultaneous optimization.

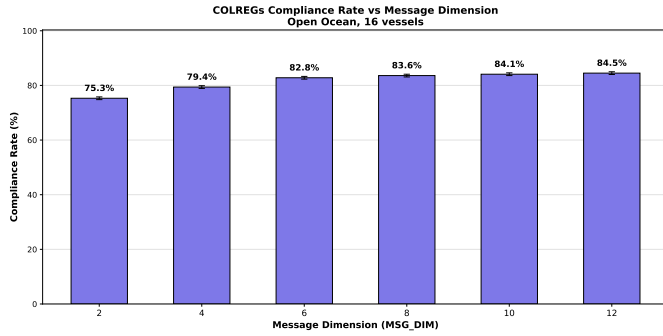


Figure 16: Message-dimensionality ablation in Open Ocean (16 vessels, 10 runs $\times$ 2,000 steps). (a) COLREGs compliance and (b) episode time as a function of message dimension $|\mathbf{m}| \in \{2, 4, 6, 8, 10, 12\}$. Both metrics saturate beyond $|\mathbf{m}| = 6$.

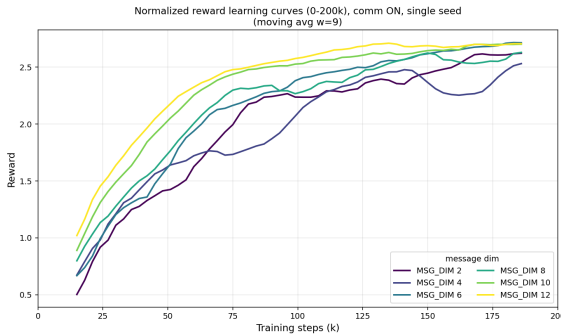


Figure 17: Normalized reward learning curves for message dimensions $|\mathbf{m}| \in \{2, \dots, 12\}$ (communication ON). Larger dimensions converge faster early in training, but all dimensions ≥ 6 reach an indistinguishable asymptote.

this system is intended to model. Dimension 6 thus offers the best trade-off between representational sufficiency and communication parsimony.

7.4. Proportion of Communicating Vessels

A practical maritime fleet is heterogeneous: not every vessel will be equipped to broadcast cooperative messages. To as-

sess how the benefit of communication depends on its adoption rate, we vary the composition of the 16-vessel fleet between fully communicating agents and receive-only agents—vessels that integrate received messages but do not transmit their own. We sweep the mix from 8 receive-only/8 communicating up to 2 receive-only/14 communicating. Fig. 18 reports the results from two complementary viewpoints.

At the fleet level (Fig. 18a), aggregate COLREGs compliance rises monotonically with the proportion of communicating vessels, from 81.3% (8 receive-only) to 84.0% (2 receive-only), confirming that broader communication adoption improves overall fleet safety. Disaggregating by agent type (Fig. 18b) reveals the mechanism: across every fleet composition, communicating agents consistently outperform receive-only agents—higher compliance (83–84% vs. 74–77%), larger mean DCPA ($+ \sim 17\%$), fewer dangerous proximity events ($- \sim 15\%$), and shorter episode times ($- \sim 14\%$). Crucially, this per-type advantage is preserved regardless of how many vessels communicate, which isolates the transmitting capability itself—rather than any confound of fleet density—as the causal driver of the performance gain. The monotonic fleet-level trend additionally indicates graceful degradation: the system does not require universal participation to function, but improves smoothly as more vessels join the cooperative protocol.

7.5. MoE Reward Architecture: Specialized vs. Shared Experts

We isolate the contribution of the per-rule expert decomposition by holding the fuzzy gating mechanism fixed—identical urgency scoring (Eq. 23) and dominant-encounter selection (Eq. ??)—and varying only the network that maps the gated situation to control behavior. The *specialized* variant is our proposed MoE, in which each COLREGs encounter type is handled by a dedicated expert sub-network; the *shared* variant routes all situations through a single network of equal total capacity. Because four specialized experts collectively contain roughly four times the parameters of a single expert, a naive comparison at fixed width would place the specialized variant at a capacity and sample-efficiency disadvantage. We therefore

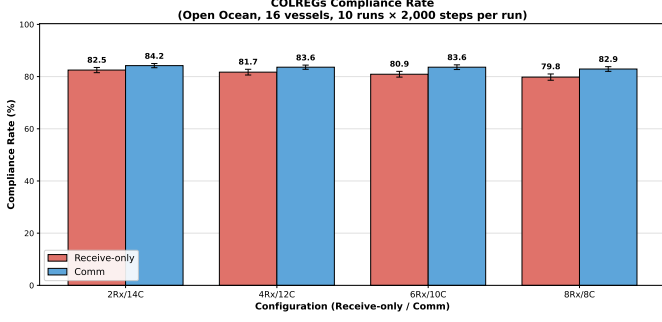


Figure 18: Communicating-vessel proportion ablation in Open Ocean (16 vessels, 10 runs $\times$ 2,000 steps). (a) Fleet-averaged COLREGs compliance as the number of receive-only vessels decreases (more communicating vessels). (b) Per-type breakdown showing that communicating agents outperform receive-only agents across all fleet compositions.

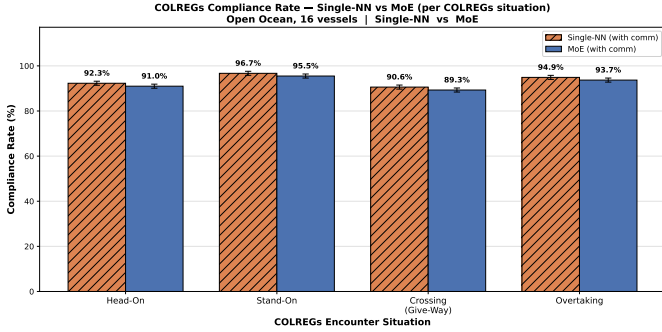


Figure 19: MoE reward-architecture ablation in Open Ocean (16 vessels, 10 runs $\times$ 2,000 steps), with the fuzzy gating held fixed and network capacity matched between variants. Specialized per-rule experts versus a single shared network of equal total parameter count, reported per COLREGs encounter type. The specialized variant yields a small, consistent gain concentrated in the hardest rules.

conduct a *capacity-matched* comparison, widening the shared network so that both variants have an equal total parameter count and are trained under an identical budget. Fig. 19 reports the comparison.

Under capacity-matched training, the specialized MoE attains a modest but consistent improvement over the shared network, with the gain concentrated in the encounter types that are hardest to satisfy—most notably Crossing Stand-On, where the rule requires holding course unless collision becomes imminent, a behavior easily diluted when a single network must also represent the opposite give-way response. This pattern is consistent with the motivation for the MoE: specializing capacity per rule prevents destructive interference between behaviorally opposite obligations, so that learning to give way does not degrade the learned stand-on policy and vice versa. We note that the absolute margin is small; the value of the specialization is therefore best understood not as a large average gain but as a structural safeguard against cross-rule interference, realized without any increase in model capacity over the shared alternative.

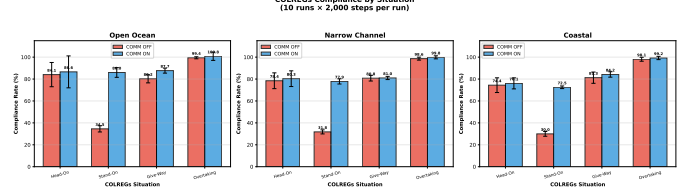


Figure 20: Curriculum ablation. COLREGs compliance by encounter type across Open Ocean, Narrow Channel, and Coastal environments for Phase 1 (communication OFF) versus Phase 2 (communication ON). Activating communication improves compliance for every rule in every environment, with the largest gain on Crossing Stand-On.

7.6. Three-Phase Curriculum

Finally, we examine the staged curriculum. The training reward curves of Fig. 7 already establish the strict ordering Phase 1 < Phase 2 < Phase 3 in asymptotic reward, demonstrating that each newly introduced objective raises performance without catastrophic forgetting of previously acquired behavior. Here we quantify the per-phase effect on task-level metrics. Fig. 20 decomposes the contribution of each curriculum stage by COLREGs situation and environment.

Activating communication (Phase 1 $\rightarrow$ Phase 2) produces the larger of the two improvements, raising compliance across all four encounter types and all three environments, with the greatest gain on the most difficult Crossing Stand-On rule. Elevating the COLREGs reward weight once cooperative awareness is already established (Phase 2 $\rightarrow$ Phase 3) yields a further, smaller, and uniformly positive refinement, together with a modest reduction in episode time. The fact that introducing communication and then sharpening the compliance incentive each improve performance—rather than destabilizing the previously learned policy—supports the central premise of the curriculum: staging the objectives allows each capability to consolidate before the next is added. By contrast, the joint-training baseline of Section 7.1, which exposes the agent to all objectives simultaneously from initialization, must contend with uninformative early messages and a strong compliance penalty before basic navigation is reliable, and converges to a lower asymptote.

7.7. Summary

As consolidated in Table 3, each contribution adds an independent and measurable improvement on top of the baseline—which, as discussed, corresponds to the design of Wang and Zhao [32] reconstructed in our environment. Multi-neighbor sum aggregation is responsible for the elimination of collisions and the bulk of the efficiency gain; a message dimension of six is the minimal sufficient channel width; broader communication adoption improves fleet-level safety monotonically while the transmitting capability itself is confirmed as the causal factor; the capacity-matched MoE provides a structural safeguard against cross-rule interference; and the staged curriculum delivers the strict Phase 1 < Phase 2 < Phase 3 performance ordering. Collectively, these results show that the full framework’s advantage over the cooperative-communication baseline is not attributable to any single component but to the conjunction of all four design choices.

8. Conclusion

A fundamental limitation of treating autonomous vessels as isolated agents is that each vessel can only react to what it can directly observe. Radar and AIS provide a partial, delayed picture of the surrounding environment — sufficient for simple scenarios, but inadequate when multiple vessels converge in confined waterways with competing right-of-way obligations. The central argument of this paper is that cooperation should not be an optional add-on but a structural property of the navigation architecture: vessels that actively share their partial observations with neighbors allow the fleet as a whole to construct a more complete situational picture than any individual agent could achieve alone.

This paper presented a cooperative MASS navigation framework built on that principle. Each vessel encodes its 371-dimensional local observation into a compact 6-dimensional latent message and broadcasts it to neighbors within the communication range. The ControlActor integrates received messages with its own observation to generate actions, effectively extending its perceptual horizon beyond the physical radar range. A Mixture-of-Experts mechanism identifies the most critical COLREGs obligation among multiple simultaneous encounters, ensuring that each vessel responds to the right rule at the right moment rather than averaging across competing signals. A three-phase curriculum introduces navigation, communication, and COLREGs compliance in sequential stages, allowing each capability to stabilize before the next is added.

Experimental results across Open Ocean, Narrow Channel, and Coastal environments with 16 vessels demonstrate the quantitative benefit of this cooperative architecture. Communication reduces average collisions per episode by 87–90%, raises COLREGs compliance from the 73–79% range to the 91–94% range, and improves goal success rates by 10–17 percentage points. A detailed per-setting analysis further shows that communication contributes in a distinct way in each environment — restoring stand-on compliance in open water, reducing close-quarters encounters under the narrow channel’s tight lateral spacing, and freeing maneuvering capacity for terrain avoidance along the coast — with the safety-margin and efficiency gains largest in the most spatially constrained settings. Rudder angle analysis confirms that cooperative agents produce smoother, lower-amplitude control commands — a direct behavioral signature of earlier encounter awareness. Scalability experiments show that these gains are maintained as fleet size grows from 4 to 16 vessels, with the communication-enabled agent holding collision rates near zero and COLREGs compliance above 94% throughout. Ablation studies further isolate the contribution of each design choice, confirming that multi-neighbor aggregation, message dimensionality, communication adoption, the Mixture-of-Experts reward, and the staged curriculum each contribute independently to the overall performance. t-SNE analysis of the learned message space reveals that the 6-dimensional messages form well-separated clusters corresponding to the four COLREGs encounter types, with interpretable gradients encoding yaw rate, speed, goal proximity, and heading — confirming that the fleet has learned a consistent

and semantically meaningful communication protocol.

Future work will investigate dynamic communication topologies that adapt bandwidth to encounter urgency, incorporate realistic AIS message latency and packet loss models, and validate the learned policies in higher-fidelity ship simulation environments. Extension to heterogeneous fleets with mixed vessel types, and integration with global route planning algorithms for end-to-end voyage optimization, represent promising directions toward practically deployable cooperative MASS navigation systems.

Acknowledgement

Nomenclature

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Table 4: Main notation for the vessel, navigation, sensing, and communication models.

Symbol	Description
<i>Agents and Sets</i>	
N	Number of vessel agents
i, j	Vessel indices
$\mathcal{I}$	Set of vessel agents, $\{1, \dots, N\}$
$\mathcal{N}_i$	Set of communication neighbors of vessel i
<i>Vessel State and Dynamics</i>	
η	Pose vector $[x, y, \psi]^\top$ in the earth-fixed frame
$\mathbf{v}$	Velocity vector $[u, v, r]^\top$ in the body-fixed frame
x, y	Vessel position in the earth-fixed frame
ψ	Heading angle
u, v, r	Surge velocity, sway velocity, and yaw rate
$\mathbf{T}(\psi)$	Rotation matrix for heading angle ψ
$\mathbf{M}$	Inertia matrix including rigid-body and added-mass terms
$\mathbf{C}(\mathbf{v})$	Coriolis and centripetal matrix
$\mathbf{D}(\mathbf{v})$	Hydrodynamic damping matrix
$\mathbf{g}(\eta)$	Hydrostatic restoring-force vector
$\mathbf{B}$	Actuator configuration matrix
$\boldsymbol{\tau}$	Control input vector
δ	Rudder angle
$v_{\max}$	Maximum vessel speed
<i>Sensing and Communication</i>	
r_{radar}	Radar detection range
r_{comm}	Inter-vessel communication range
N_{comm}	Maximum number of communication partners
d_g	Euclidean distance from the vessel to its goal
$d_{\max}$	Map diagonal used for distance normalization
θ_g	Signed angular deviation from the current heading to the goal

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- [33] PLACEHOLDER — replace with the intended control-effort/energy reference for the fuel proxy (per-step thrust–rudder norm).

Table 5: Main notation for observation, message passing, COLREGs assessment, and learning.

Symbol	Description
<i>State, Observation, and Action Spaces</i>	
$\mathcal{S}$	Global state space
$\mathcal{A}$	Joint action space
$\mathcal{O}$	Observation space
o_i	Local observation of vessel i
$\mathbf{a}_i$	Control action of vessel i
<i>Observation Components</i>	
$\mathbf{O}_{\text{radar}}$	Radar observation vector
$\mathbf{O}_{\text{goal}}$	Goal observation in ego-centric polar coordinates
$\mathbf{O}_{\text{self}}$	Self-state observation vector
$\mathbf{O}_{\text{COLREGs}}$	COLREGs encounter-type vector
<i>Message Passing</i>	
$\mathbf{m}_i$	Latent message generated by vessel i
$\bar{\mathbf{m}}_i$	Aggregated message received by vessel i from its communication neighbors
<i>COLREGs Assessment</i>	
c^*	Dominant COLREGs encounter type
g^j	Urgency score associated with neighboring vessel j
j^*	Neighbor with the highest urgency score, $\arg \max_j g^j$
$\mu_d, \mu_\tau, \mu_\delta$	Fuzzy membership functions for distance, TCPA, and DCPA
w_d, w_τ, w_δ	Weights of the fuzzy urgency components
T_0, D_0	Reference time and distance for the membership functions
β_c	Encounter-dependent urgency scaling factor
ϕ_{ij}	Signed bearing from vessel i to vessel j relative to the heading of vessel i
$g_{\min}$	Urgency threshold for declaring an active encounter
<i>Reward</i>	
R_{total}	Total reward at each time step
R_{goal}	Goal-arrival reward
$R_{\text{collision}}$	Collision penalty
R_{spinning}	Spinning penalty
R_{time}	Time-step penalty
R_{forward}	Forward-progress reward
R_{COLREGs}	COLREGs compliance reward
r_{c^*}	Compliance score associated with the active rule c^*
α_c	COLREGs reward coefficient
<i>Training</i>	
$\mathcal{L}_{\text{total}}$	Total PPO training loss
γ	Discount factor
λ	GAE parameter
ϵ	PPO clipping parameter
α_v	Value-loss coefficient
α_e	Entropy-bonus coefficient
α_{cls}	COLREGs classifier-loss weight

Table 6: System and communication parameters.

Parameter	Symbol	Value
Radar detection range	r_{radar}	60 m
Communication range	r_{comm}	90 m
Maximum communication partners	N_{comm}	4
Maximum vessel speed	v_{max}	—
Map diagonal	d_{max}	200 m
Observation dimension	—	371
Radar observation dimension	$ \mathbf{O}_{\text{radar}} $	360
Goal observation dimension	$ \mathbf{O}_{\text{goal}} $	2
Self-state dimension	$ \mathbf{O}_{\text{self}} $	4
COLREGs observation dimension	$ \mathbf{O}_{\text{COLREGs}} $	5
Message dimension	$ \mathbf{m}_i $	6
Reference time	T_0	60 s
Reference distance	D_0	50 m
Urgency weights	(w_d, w_τ, w_δ)	(0.3, 0.4, 0.3)

Table 7: Encounter-dependent urgency scaling factors.

Encounter type	β_c
Head-On	2.0
Give-Way	1.5
Stand-On	1.3
Overtaking	1.2

Table 8: PPO training hyperparameters.

Hyperparameter	Symbol	Value
Discount factor	γ	0.99
GAE lambda	λ	0.95
Learning rate	α	3×10^{-4}
Batch size	B	2,048
PPO epochs per update	K	4
PPO clipping	ε	0.2
Entropy bonus coefficient	α_e	0.01
Value loss coefficient	α_v	0.5
COLREGs reward coefficient (Phase 1)	α_c	0.30
COLREGs reward coefficient (Phase 3)	α_c	0.45
COLREGs classifier loss weight	α_{cls}	0.10
Gradient clipping norm	—	0.5
Phase 1 training steps	—	4.07×10^6
Total training steps	—	30×10^6
Observation dimension	—	371
Message dimension	$ \mathbf{m}_i $	6
Frame stacking	F	3
Conv. kernel sizes	—	5, 3
Conv. channels	—	32, 32
FC hidden units	—	$256 \rightarrow 128 \rightarrow 64$